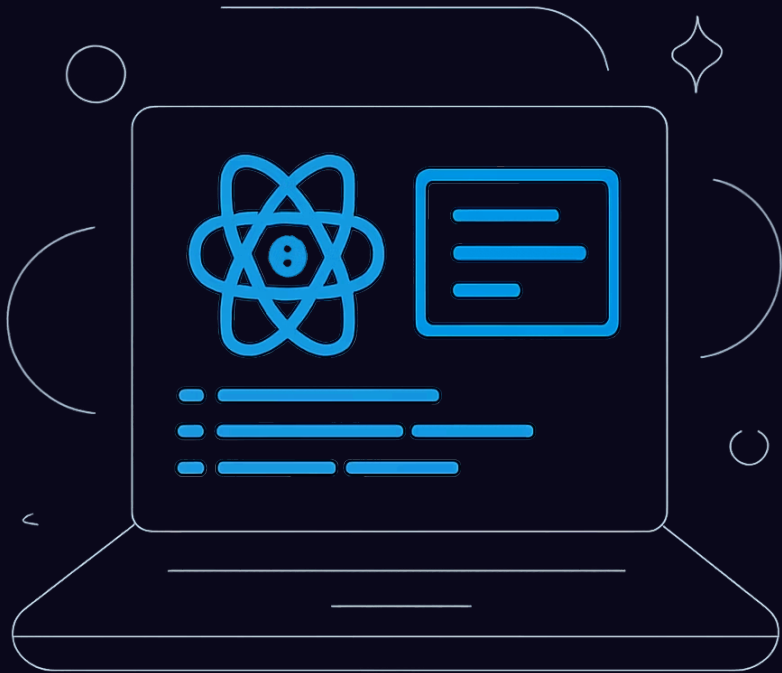




React JSX, Props & Component Communication

Today You Will Learn

- What is JSX?
- JSX Rules
- Expressions in JSX
- Props
- Default Props
- Children Props
- Component Reusability
- Key Prop
- Rendering Lists
- Mini Project using Bootstrap Cards

📌 **Learning Outcome:** By the end of today's session, you will be able to build reusable React components and pass data between components using Props.

What is JSX?

JSX (JavaScript XML)

JSX allows us to write HTML-like code inside JavaScript.

Instead of writing:

```
React.createElement("h1", null, "Hello React");
```

We write:

```
function App() { return <h1>Hello React</h1>; }
```

Why JSX?

Easy to read

Easy to write

Looks like HTML

Faster UI development

Converted into JavaScript by Babel

Real World Example

Almost every React application uses JSX because it makes UI development simple and maintainable.

JSX Rules

Rule 1: Return Only One Parent Element

✗ Wrong

```
<h1>Hello</h1>  
<p>Welcome</p>
```

✓ Correct

```
<>  
  <h1>Hello</h1>  
  <p>Welcome</p>  
</>
```

Rule 2

Use **className** instead of **class**

Rule 3

Close every tag.

Rule 4

JavaScript goes inside curly braces.

```
{name}
```



Expressions in JSX

JavaScript expressions can be used inside JSX.

```
const name = "Rahul";  
  
<h1>Hello {name}</h1>
```

Examples

```
{10 + 20}  
{name.toUpperCase()}  
{price * quantity}
```

Conditional Rendering

```
{age >= 18 ? "Eligible" : "Not Eligible"}
```

Note

You cannot directly write:

- if statements
- for loops

Instead use:

- Ternary Operator
- map()

Props (Component Communication)

What are Props?

Props are used to pass data from a Parent Component to a Child Component.

Parent Component

Child Component

```
function EmployeeCard(props) {  
  return (  
    <>  
      <h2>{props.name}</h2>  
      <p>{props.role}</p>  
    </>  
  );  
}
```

Remember

Props are Read Only.

Parent sends data.

Child receives data.

Default Props & Children Props

Default Props

Provide a default value when no prop is passed.

```
function Employee({ name = "Unknown" }) {  
  return <h2>{name}</h2>;  
}
```

Children Props

Anything between opening and closing tags becomes **children**.

```
<Card>  
  <h2>React Course</h2>  
  <p>Learn React</p>  
</Card>  
  
function Card({ children }) {  
  return (  
    <div>{children}</div>  
  );  
}
```

Used for



Layouts



Reusable wrappers



Dashboards



Cards



Modals

Component Reusability

Instead of repeating HTML, create reusable components.

Benefits



Less Code



Easy Maintenance



Reusable UI



Better Readability



Faster Development



Software Engineering Principle: Build Once, Reuse Everywhere.

Rendering Lists & Key Prop

Rendering Multiple Components

```
const employees = [  
  { id:1, name:"Rahul" },  
  { id:2, name:"Priya" }  
];  
  
employees.map(employee => <EmployeeCard key={employee.id} {...employee} />)
```

Why Key?

React uses keys to identify each component efficiently.

✓ Good

```
key={employee.id}
```

Avoid

```
key={index}
```

Use index only when the list never changes.

Mini Project: Bootstrap Employee Cards using Props

Objective

Create reusable Bootstrap cards by passing employee details through Props.

Install Bootstrap

```
npm install bootstrap
```

Import in main.jsx

```
import "bootstrap/dist/css/bootstrap.min.css";
```

EmployeeCard Component

```
function EmployeeCard({ name, role, experience, image }) {  
  return (  
    <div className="card">  
      <img src={image} className="card-img-top" />  
      <div className="card-body">  
        <h4 className="card-title">{name}</h4>  
        <p className="card-text">  
          <b>Role:</b> {role}<br />  
          <b>Experience:</b> {experience}  
        </p>  
        <a href="#" className="btn btn-primary">View Profile</a>  
      </div>  
    </div>  
  );  
}
```

App.jsx

Learning Outcome

→ Bootstrap Cards

→ Component Reusability

→ Props

→ Passing Dynamic Data

Homework Assignment

1 Homework 1: Student List using Props Create a reusable StudentCard component. Each card should display: • Student Photo • Student Name • Roll Number • Course • College • Percentage Render 10 student cards using <code>map()</code>.	2 Homework 2: Use Default Props If a student's percentage is not available, display: Percentage: Not Available
3 Homework 3: Use Children Props Create a reusable Card component. Example:	4 Homework 4: Build Product Cards Create a reusable ProductCard component. Display: • Product Image • Product Name • Price • Rating ★ • Buy Now Button Render at least 8 products using <code>map()</code> and assign a unique key to each card.

Submission Checklist

✓	✓
✓ JSX Rules followed	✓ Used Props correctly
✓	👍
✓ Implemented Default Props	✓ Used Children Props
📄	✓
✓ Rendered lists using <code>map()</code>	✓ Added unique key for each item
✓	✓
✓ Used Bootstrap Cards	✓ Created reusable components

☐ **Expected Outcome:** Students should be able to build reusable React components, pass data through props, render dynamic lists, and create professional Bootstrap card layouts similar to real-world React applications.